

QMST5336: ANALYTICS

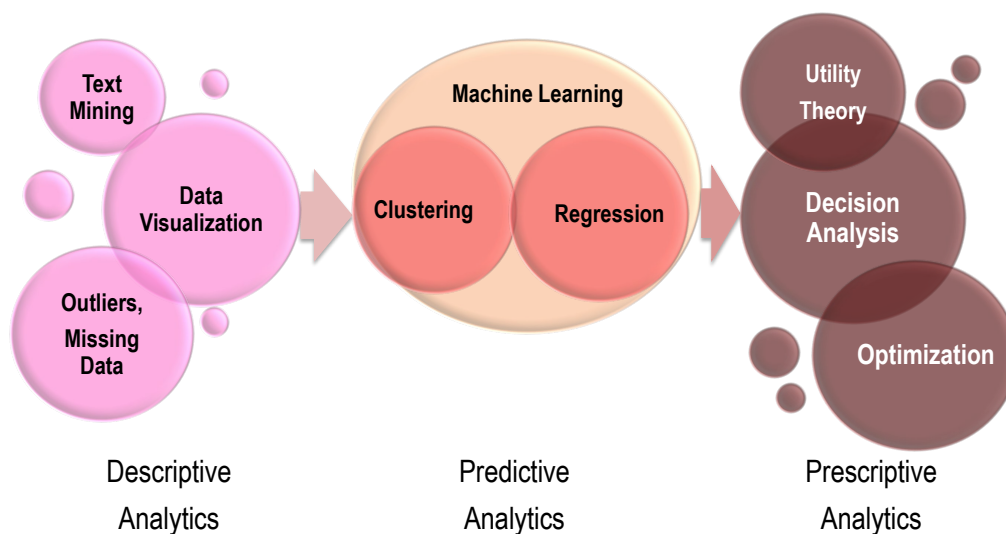
Lecture 10: Payoff Table & Utility Theory & Simulation

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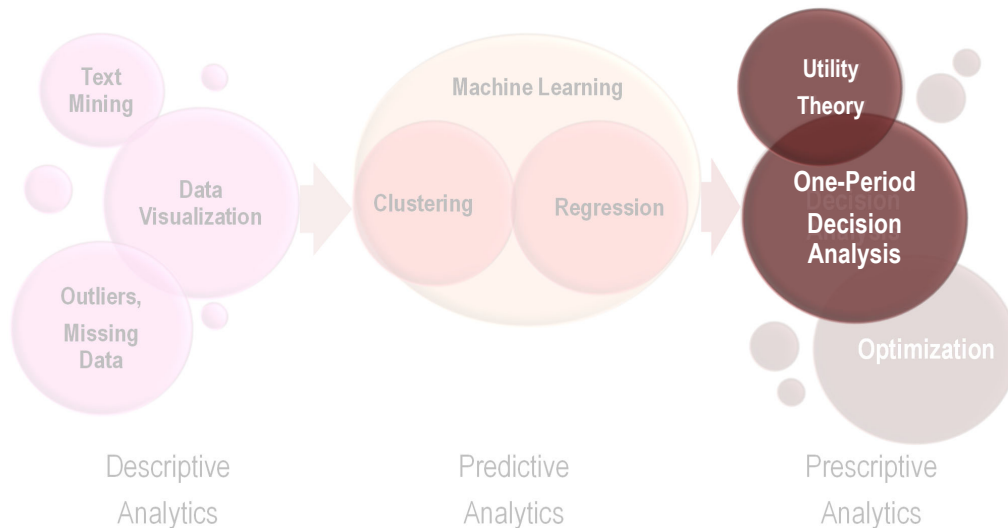
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Our Problem and Roadmap

An investment property in Austin: **Airbnb** versus **Rental**?



Roadmap: Where We Are



Case: Decide A Renovation Plan

Because of an outdated layout in the house located in north Austin, the proper price is a \$200 per night. The expected occupation rate is 120 days if the local tourism market is hot. In the case of a normal market, the expected annual occupation becomes 100 days; whereas a cold market can make the occupation rate decrease to 70 days across 3 years.

The market conditions are more or less the same in the following 3 years. The economy outlook suggests a probability of 0.35 for a hot market and 0.5 for a normal one. The chance of a cold market is 0.15.

Question 1: Which renovation plan should I choose considering my return in the next 3 years?

Case: Decide A Renovation Plan C'td

The company specialized in renovation offers 3 types of plans:

Neat & Tide plan costs \$10,000 and will increase the occupation rate by 10% for all markets. The daily price after renovation will be \$210.

Modern plan costs \$22,000 and can increase the occupation rate by 20% regardless of market environments. The daily price after renovation will become \$250.

Luxury plan costs \$50,000 and makes the property be positioned in a niche market, where the daily fare is \$400. However, the occupation rates are estimated to be 135, 85 and 40 days in a hot, normal and cold market, respectively.

Question 1: Which renovation plan should I choose considering my return in the next 3 years?

Case: A Bit Calculation

Let's **calculate** the increase (decrease) **marginal return** in the expected return of the next 3 years, **compared with status quo**.

Status Quo: return = daily rate $\times$ expected occupied days in 3 years, for different markets.

$$\begin{aligned} 200 \times 120 \times 3 &= 72,000\$ && \text{if a hot market} \\ 200 \times 100 \times 3 &= 60,000\$ && \text{if a normal market} \\ 200 \times 70 \times 3 &= 42,000\$ && \text{if a cold market} \end{aligned}$$

Renovation Plans: marginal return = profit – cost – return of no renovation.

Neat & Tide plan costs \$10,000 and will increase the occupation rate by 10% for all markets. The proper price after renovation will be \$210 per night.

$$\begin{aligned} 210 \times 120(1.1) \times 3 &= 83,160\$ \\ 83,160 - 10,000 - 72,000 &= 1,160\$ && \text{if a hot market} \\ 210 \times 100(1.1) \times 3 &= 69,300\$ \\ 69,300 - 10,000 - 60,000 &= -700\$ && \text{if a normal market} \\ 210 \times 70(1.1) \times 3 &= 48,510\$ \\ 48,510 - 10,000 - 42,000 &= -3,490\$ && \text{if a cold market} \end{aligned}$$

Modern plan costs \$20,000 and can increase the occupation rate by 20%

Case: Present the Information In A Better Way

		Hot	Normal	Cold
Options	Status Quo	0	0	0
	Neat & Tide	1.2	-.7	-3.5
	Modern	14	8	-1
	Luxury	40	12	-44
		in the unit of thousand \$		

Decision Theory

Necessary components of decision analysis:

1. A set of possible decisions (acts) $D_1, D_2, D_3, \dots, D_m$.
2. A set of possible outcomes (events) $E_1, E_2, E_3, \dots, E_n$. The set of events must be pairwise mutually exclusive and collectively exhaustive.
3. A set of payoffs R_{ij} associated with each decision-event combination.
Thus, $R_{ij} = f(D_i, E_j)$
4. A set of prior probabilities for the events: $P(E_1), P(E_2), \dots, P(E_n)$.

Payoff Tables

For a single-stage decision problem, we summarize the above information in a payoff table.

		EVENTS			
		E_1	E_2		E_n
DECISIONS	D_1	R_{11}	R_{12}		R_{1n}
	D_2	R_{21}	R_{22}		R_{2n}
	$\vdots$	$\vdots$	$\vdots$		$\vdots$
	D_m	R_{m1}	R_{m2}		R_{mn}

In general, the expression R_{ij} refers to the payoff associated with decision D_i and event E_j .

Case: Events & Decisions

Three **events** are possible:

1. E_1 : Market is hot.
2. E_2 : Normal Market.
3. E_3 : Market will cool down.

In the meantime, we have to decide among the following four **decisions**:

1. D_1 : No renovation.
2. D_2 : Neat & Tide plan
3. D_3 : Modern plan.
4. D_4 : Luxury plan.

Case: Payoff Table

The payoff table (\$1,000) is as follows:

		EVENTS		
		E_1	E_2	E_3
DECISIONS	D_1	0	0	0
	D_2	1.2	-0.7	-3.5
	D_3	14	8	-1
	D_4	40	12	-44

Inadmissible Decisions

Decision D_i dominates decision D_j if:

- $R_{ik} \geq R_{jk}$ for all $k = 1, 2, \dots, n$;
- $R_{ik} > R_{jk}$ for at least one value of k .

When D_j is an inadmissible event, it can be eliminated from the payoff table. In our case, D_2 is inadmissible (dominated by D_3). Therefore eliminate D_2 to obtain the reduced payoff table:

		EVENTS		
		E_1	E_2	E_3
DECISIONS	D_1	0	0	0
	D_3	14	8	-1
	D_4	40	12	-44

Decision criteria without probabilities

MAXIMIN PAYOFF CRITERION The maximin decision is obtained by selecting the minimum payoff for each decision (row) and then choosing the decision corresponding to the maximum of the minimum payoffs.

	E_1	E_2	E_3	<u>Min</u>
D_1	0	0	0	$0 < \text{---Max}$
D_3	14	8	-1	-1
D_4	40	12	-44	-44

The maximin decision is D_1 : status quo (i.e., do nothing)

It is a conservative, pessimistic decision. It exhibits risk averse behavior.

Decision criteria without probabilities

MAXIMAX PAYOFF CRITERION The maximax decision is obtained by selecting the maximum payoff for each decision (row) and then choosing the decision corresponding to the maximum of the maximum payoffs.

	E_1	E_2	E_3	<u>Max</u>
D_1	0	0	0	0
D_3	14	8	-1	14
D_4	40	12	-44	$40 < \text{---Max}$

The maximax decision is D_4 : renovate using a luxury plan.

It is an aggressive, optimistic decision. It exhibits risk seeking behavior.

Decision criteria without probabilities

MINIMAX OPPORTUNITY LOSS (REGRET) CRITERION The minimax O.L. decision is obtained by selecting the maximum O.L. for each decision (row) and then choosing the decision corresponding to the minimum of the maximum O.L.s.

	E_1	E_2	E_3	<u>Max</u>
D_1	0	0	0	0 ← Min
D_2	14	8	-1	14
D_4	40	12	-44	40

The minimax O.L. decision is D_1 : status quo (i.e., do nothing).

Note that it is neither overly aggressive nor overly conservative.

Decision Criteria with Probabilities

So far we have only considered decision criteria where no probabilities are available. This is often referred to as decision making under uncertainty.

When probabilities are incorporated into the decision making process, we refer to decision making under risk.

In our case, we now introduce the probabilities:

$$P(E_1) = 0.35, \quad P(E_2) = 0.5, \quad P(E_3) = 0.15$$

Calculating Expected Values

Consider the case where *Decision 3* is made. In that case there are three possible payoffs, namely 16 (probability 0.35), 10 (probability 0.5), and 1 (probability 0.15).

The expected value calculation is shown in the table below:

Event	Payoff(X)	Probability P(X)	X*P(X)
E_1	14	0.35	4.9
E_2	8	0.5	4
E_3	-1	0.15	-0.15

$$8.75 = \sum X * P(X)$$

Expected Monetary Value (EMV)

In decision modeling where the payoffs are the monetary values associated with decision-event combinations, we refer to the expected values as "expected monetary values" or EMVs.

BAYES' Decision Criterion

MAXIMUM EXPECTED MONETARY VALUE Calculate the expected monetary value (EMV) for each decision and choose the decision corresponding to the maximum EMV.

	E_1	E_2	E_3	EMV
D_1	0	0	0	0
D_3	14	8	-1	8.75
D_4	40	12	-44	13.4 < —Max
$P(E_i)$	0.35	0.5	0.15	

D_4 is the maximum EMV decision.

Perfect information

How much would perfect information be worth to our decision?

We calculate the Expected Payoff with Perfect Information (EPPI) as follows:

Perfect Information	Response	Payoff(X)	Probability P(X)	X*P(X)
E_1	D_4	40	0.35	14
E_2	D_4	12	0.5	6
E_3	D_1	0	0.15	0

20=EPPI

Then the Expected Value of Perfect Information (**EVPI**) is calculated as:

$$EVPI = EPPI - EMV$$

In the example: $EVPI = 20 - 13.4 = 6.6$ (i.e. \$6,600)

Utility Theory

Assessing the Decision Maker's (D.M.'s) Utility (Preference) for Money



Example 1: House Insurance

This example illustrates the need to assess the D.M.'s utility for money.

E_1 =Fire $P(E_1)$ =.002 D_1 =Buy insurance
 E_2 =No fire $P(E_2)$ =.998 D_2 =Don't buy insurance

	E_1	E_2	
D_1	-100	-100	-100
D_2	-40000	0	-80
$P(E_i)$	.002	.998	

Optimal EMV decision

The DM will not buy insurance because the EMV for not buying is -\$80, whereas the EMV for buying is -\$100.

Since the rationale for buying insurance is to avoid the catastrophe loss of \$40,000, the EMV decision model fails to account for this risk avoidance.

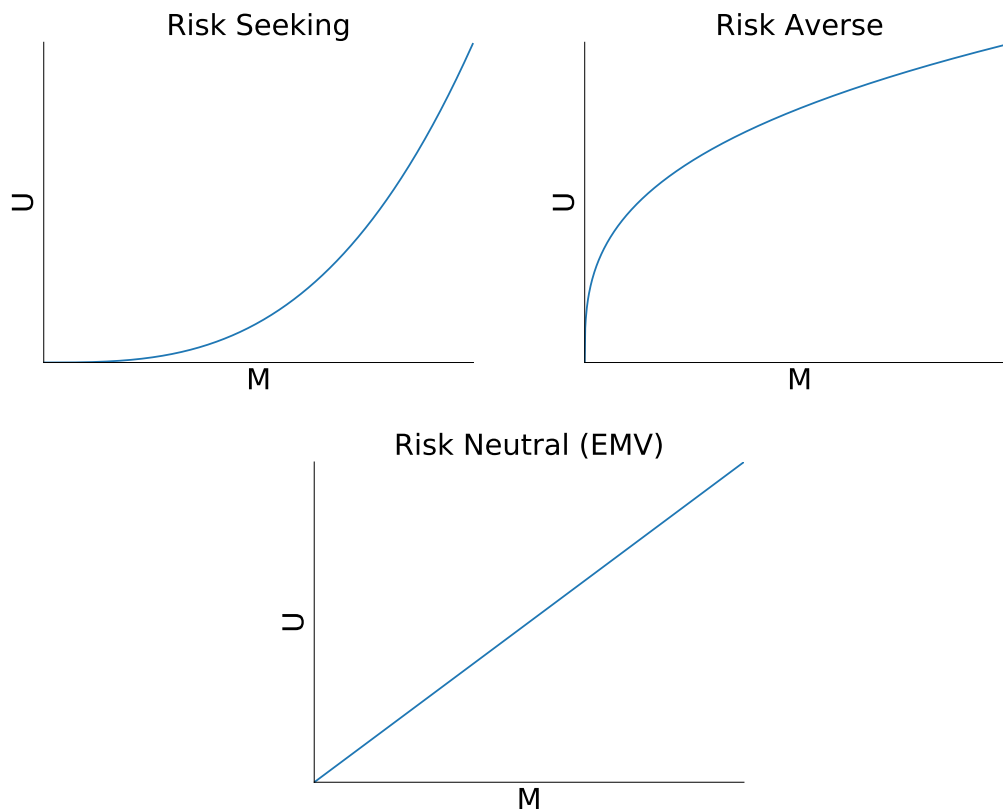
Assumption in Utility Theory

- 1 For any pair of outcomes E_1 and E_2 , the D.M. can decide if he/she prefers E_1 to E_2 , E_2 to E_1 , or likes them both equally
- 2 If E_1 is preferred to E_2 , and E_2 is preferred to E_3 , then E_1 is preferred to E_3 .
- 3 For every gamble having worst and best payoffs R_1 and R_2 , the D.M. can choose a value between R_1 and R_2 , which is equivalent to the gamble. This value is called the **certainty equivalent (CE)** of the gamble.

Utilities may be specified as:

- A function/curve.
- A set of gambles with C.E.'s.

Utility Functions



Example 2: Buy or Rent Equipment

A contractor wants to decide whether to rent or buy equipment in anticipation of a contract which he hopes to get next month. He has established the following payoff table for his decision problem (in \$1,000):

	Get Contract	Fail to get Contract
Buy	120	-40
Rent	50	0

Since only one other contractor has submitted a bid he estimates that he has a 50-50 chance of getting the contract.

$$\text{EMV}(\text{Buy}) = 120(0.5) + (-40)(0.5) = 40.$$

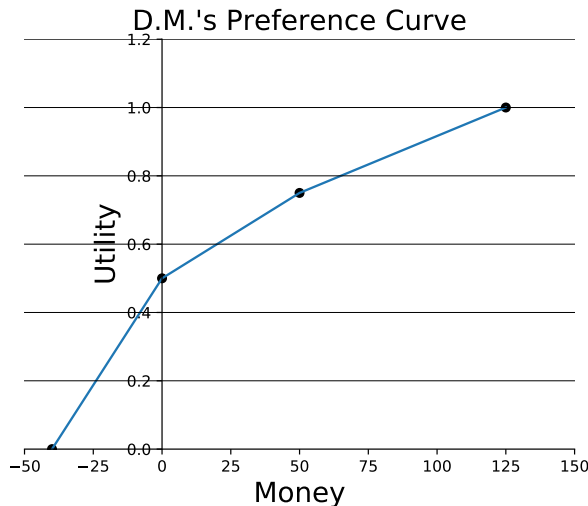
$$\text{EMV}(\text{Rent}) = 50(0.5) + 0(0.5) = 25.$$

Optimal decision: Buy the equipment

A Utility Function for the DM

Suppose the contractor's utility can be specified by a utility function

$$u(M) = \sqrt{\frac{(M + 40)}{160}}$$



Money	Utility
-40	0
0	.5
50	.75
120	1

$$EU(\text{Buy}) = 1(0.5) + 0(0.5) = 0.5$$

$$EU(\text{Rent}) = .75(0.5) + .5(0.5) = 0.625$$

Optimal decision: Rent the equipment

We see that the risk averse nature of the utility function has caused the DM to change from the aggressive "Buy" decision to the conservative "Rent" decision.

Formula for Computing CEs

GAMBLE

Payoff

Probability

R_1

p_1

R_2

p_2

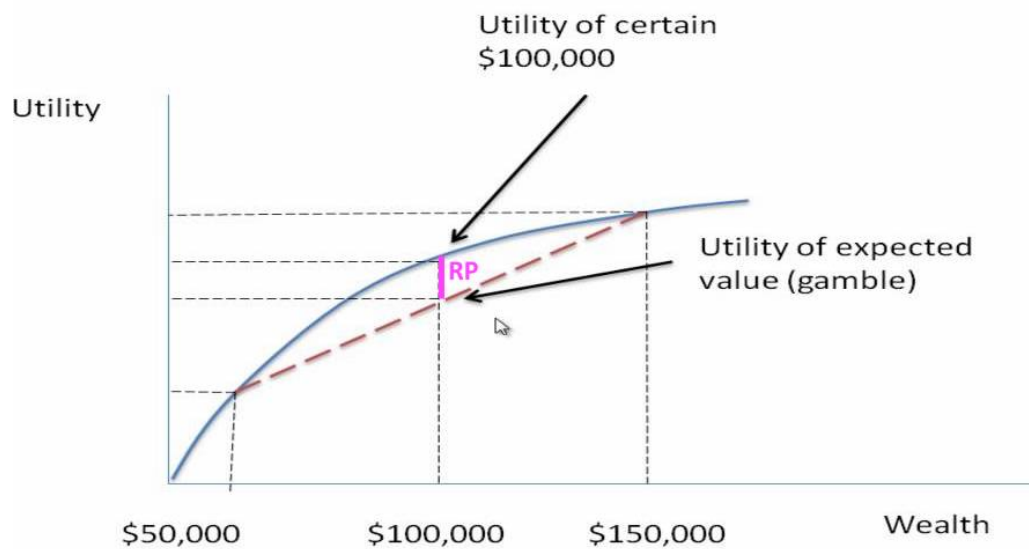
$$U(\text{CE}) = U(R_1)p_1 + U(R_2)p_2$$

Rationale The Decision Maker is indifferent between the gamble and the Certainty Equivalent (i.e. he/she will accept the CE as being equivalent to the gamble). Hence they both have the same utility for the D.M. The formula shows that the utility of the CE is the same as the expected utility of the gamble.

Risk Premium

$$\text{Risk Premium} = \text{EMV} - \text{CE}$$

- Risk Premium $> 0 \implies$ Risk Averse
- Risk Premium $= 0 \implies$ Risk Neutral
- Risk Premium $< 0 \implies$ Risk Seeking



Simulation



Simulation Introduction

A small bakery makes a specialty item, Pumpkin Pies, which has the following distribution of demand:

Demand:D	Probability:P(D)
0	.10
1	.30
2	.40
3	.20

Pies are baked during the night and no further baking is possible during the day, so that if demand exceeds the number of units in stock each morning, the sales are lost.

It costs \$2.50 to bake a pie, and each unit sells for \$5.50. Any units that are not sold during the day will be bought by a bakery employee at the discounted price of \$1.00.

Evaluate each of the policies: Stock 1 daily, Stock 2 daily, Stock 3 daily.

Question: What is the optimal number of pies to stock?

Payoff Tables

First we note that this problem can be solved as a payoff table model as follows:

Cost Price = \$2.50

Selling Price = \$5.50

Salvage Value = \$1.00

Calculating the payoffs for each stock/demand combination we obtain the following payoff table:

		Demand				
		0	1	2	3	EMV
stock	0	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
	1	-\$1.50	\$3.00	\$3.00	\$3.00	\$2.55
	2	-\$3.00	\$1.50	\$6.00	\$6.00	\$3.75 <—Optimal Profit
	3	-\$4.50	\$0.00	\$4.50	\$9.00	\$3.15
Prob:		0.10	0.30	0.40	0.20	

The decision to stock 2 yields the highest expected profit (expected monetary value), therefore the optimal decision is to stock 2 pies daily. This results in an average daily profit of \$3.75.

Calculation for Payoff Tables

For example, if you stock 3 and demand is for 2, your profit will be calculated as:

$$\text{Revenue} = 2 * \$5.50 = \$11.00$$

$$\text{Salvage} = 1 * \$1.00 = \$1.00$$

$$\text{Cost} = 3 * \$2.50 = \$7.50$$

$$\text{Profit} = \$11.00 + \$1.00 - \$7.50 = \$4.50$$

For example, if you stock 2 and demand is for 3 your profit is given by:

$$\text{Revenue} = 2 * \$5.50 = \$11.00$$

(notice, even though demand is for 3 you only have 2 to sell)

$$\text{Cost} = 2 * \$2.50 = \$5.00$$

$$\text{Profit} = \$11.00 - \$5.00 = \$6.00$$

Decision Using Simulation

Simulation is a way to model uncertainty.

Simulate 10 days of sales using the following random numbers:

Day	1	2	3	4	5	6	7	8	9	10
RN	97	61	11	50	86	25	06	60	80	19

Next, let us see how we can simulate the bakery's pie sales problem.

The probability distribution is:

Demand:D	Probability:P(D)
0	.10
1	.30
2	.40
3	.20

To simulate this distribution we will generate random numbers between 00 and 99, inclusive.

Simulate Uncertain Demands

The first 10 numbers 00-09 will be assigned to the value $D = 0$. This implies that 10 of the 100 2-digit numbers will correspond to $D = 0$. That is a probability of 10% or 0.10.

We will then assign the next 30 numbers, namely 10-39, to $D = 1$. This will guarantee that 30 out of 100 or 30% of the 2-digit random numbers will correspond to $D = 1$. We continue this process until we have constructed the following table:

D	P(D)	Range of Random Numbers
0	.10	00-09
1	.30	10-39
2	.40	40-79
3	.20	80-99

We now see how to simulate daily demand: we generate a set of random numbers and determine the demand associated with each number. For example, a $RN = 48$ simulates a demand for 2 pies, since 48 lies in the range 40-79.

Terminology on Simulation Table

In the simulation tables you will see the following terms:

Overage refers to the number of units of overstocking

For example, if $Q = 2$ and $D = 1$ then $\text{overage} = 2 - 1 = 1$. In other words, you have 1 unit in stock for which there is no demand, i.e. you are overstocked by 1 unit.

Underage refers to the number of units of understocking

For example, if $Q = 1$ and $D = 3$ then $\text{underage} = 3 - 1 = 2$. In other words, you have demand for 2 units that are not in stock, i.e. you are understocked by 2 units.

Cost of Lost Sales refers to the opportunity cost associated with each unit of underage.

If you have demand that you cannot fill, due to being understocked, you lose the potential profit for that lost sale. In this example the cost of underage is \$3 per unit ($\text{Selling Price} - \text{Cost Price} = \$5.50 - \$2.50 = \3.00)

Simulation Table: stock quantities $Q = 1$

Q=1							
Day	Random Number	Demand	Sales	Underage	Overage	Profit	Cost of Lost sales
1	97	3	1	2	0	\$3.00	\$6.00
2	61	2	1	1	0	\$3.00	\$3.00
3	11	1	1	0	0	\$3.00	\$0.00
4	50	2	1	1	0	\$3.00	\$3.00
5	86	3	1	2	0	\$3.00	\$6.00
6	25	1	1	0	0	\$3.00	\$0.00
7	06	0	0	0	1	-\$1.50	\$0.00
8	60	2	1	1	0	\$3.00	\$3.00
9	80	3	1	2	0	\$3.00	\$6.00
10	19	3	1	1	0	\$3.00	\$0.00
Totals		18	9	9	1	\$25.50	\$27.00
Average		1.8	0.9	0.9	0.1	\$2.55	\$2.70

Simulation Table: stock quantities $Q = 1$ C'td

For $Q = 1$, average daily profit is \$2.55.

Note: in this simulation, Sales = Underage.

Note: The daily cost of lost sales is \$2.70, which is greater than the daily profit. This suggests that we should increase the order quantity.

Note: Demand = Sales + Underage (i.e. $18 = 9 + 9$)

Simulation Table: stock quantities $Q = 2$

Q=2							
Day	Random Number	Demand	Sales	Underage	Overage	Profit	Cost of Lost sales
1	97	3	2	1	0	\$6.00	\$3.00
2	61	2	2	0	0	\$6.00	\$0.00
3	11	1	1	0	1	\$1.50	\$0.00
4	50	2	2	0	0	\$6.00	\$0.00
5	86	3	2	1	0	\$6.00	\$3.00
6	25	1	1	0	1	\$1.50	\$0.00
7	06	0	0	0	2	-\$3.00	\$0.00
8	60	2	2	0	0	\$6.00	\$0.00
9	80	3	2	1	0	\$6.00	\$3.00
10	19	1	1	0	1	\$1.50	\$0.00
Totals		18	3	5	1	\$37.50	\$9.00
Average		1.8	1.5	0.3	0.5	\$3.75	\$0.90

Simulation Table: stock quantities $Q = 2$ C'td

For $Q = 2$, average daily profit is \$3.75.

Note: in this simulation, $\text{Sales} > \text{Underage}$.

Note: The daily cost of lost sales is \$0.90

Note: $\text{Demand} = \text{Sales} + \text{Underage}$ (i.e. $18 = 15 + 3$)

Simulation Table: stock quantities $Q = 3$

Q=3							
Day	Random Number	Demand	Sales	Underage	Overage	Profit	Cost of Lost sales
1	97	3	3	0	0	\$9.00	\$0.00
2	61	2	2	0	1	\$4.50	\$0.00
3	11	1	1	0	2	\$0.00	\$0.00
4	50	2	2	0	1	\$4.50	\$0.00
5	86	3	3	0	0	\$9.00	\$0.00
6	25	1	1	0	2	\$0.00	\$0.00
7	06	0	0	0	3	-\$4.50	\$0.00
8	60	2	2	0	1	\$4.50	\$0.00
9	80	3	3	0	0	\$9.00	\$0.00
10	19	1	1	0	2	\$0.00	\$0.00
Totals		18	18	0	12	\$36.00	\$0.00
Average		1.8	1.8	0	1.2	\$3.60	\$0.00

Simulation Table: stock quantities $Q = 3$

For $Q = 3$, average daily profit is \$3.60.

Note: in this simulation, we have completely eliminated underage.

Note: The daily cost of lost sales is \$0.00

Note: Demand = Sales + Underage (i.e. $18 = 18 + 0$). In this case we are selling all units demanded.

Note: in each of the three simulations the first 3 columns are the same because we want to compare the effect of different order quantities on the same ten days of demand.

Simulation Summary

Quantity	Sales	Underage	Overage	Cost of Lost Sales	Daily Profit
Q=1	9	9	1	\$2.70	\$2.55
Q=2	15	3	5	\$0.90	\$3.75
Q=3	18	0	12	\$0.00	\$3.60

Conclusion

Optimal order quantity $Q^* = 2$. This results in an expected daily profit of \$3.75.

Payoff Table versus Simulation

Comparison of Payoff Table (exact expected values) and Simulation (sample expected values)

Q	Payoff Table	Simulation
1	\$2.55	\$2.55
2	\$3.75	\$3.75
3	\$3.15	\$3.60

We observe that our simulation of 10 days of operation is very accurate, yielding the exact values for $Q = 1$ and $Q = 2$, but overestimating the value when $Q = 3$.

In general we do not expect the simulation results to be this accurate, but as the number of days simulated increases, the simulated values will **converge** to the expected values.

More Complicated Example

Cars arrive at one Texas-Nuevo Leon border periodically throughout the day. Cars are randomly selected and subjected to a search. Thirty-five percent of all cars are searched. Only one customs official works a small rural border crossing and all cars must pass by him or her. The length of the search is also variable. The border security agency wants to study the one-hour period from midnight to 1:00 a.m. The following probability distributions described the number of cars arriving during a typical five-minute period, and the length of time the customs agent spends searching the car, rounded to the nearest two minutes.

No. of Arrivals	Probability
0	0.10
1	0.25
2	0.45
3	0.20

Search Time(minutes)	Probability
2	0.40
4	0.25
6	0.20
8	0.10
10	0.05

Questions

- What percentage of cars that arrive between 12:00 and 1:00 a.m are searched?
- What percentage of the time is the customs agent busy during the period from 12:00 and 1:00 a.m?

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43 / 51

What Are Uncertain?

Step1: There are THREE random events, we need to determine the random number assignment SCHEME for each of them according to their probability distributions.

Arrivals	P	Assigned RN
0	0.10	0-09
1	0.25	10-34
2	0.45	35-79
3	0.20	80-99

Search Time(minutes)	P	Assigned RN
2	0.40	0-39
4	0.25	40-64
6	0.20	65-84
8	0.10	85-94
10	0.05	95-99

What Are Uncertain?

Event	P	Assigned RN
Search	0.35	0-34
No search	0.65	35-99

- 1 Simulate the arrival cars between midnight and 1:00 a.m. using a series of random numbers
- 2 Use a 2nd series of random numbers for the decision to search or not to search each car that arrives. for example,
15 94 33 46 19 48 52 96 80 54 23 86 69 44 11 32 41 78
- 3 Use another series of random numbers to simulate search time for each car that is searched: for example, 42 02 47 56 67 31 17

Independent Random Numbers

These THREE series of random numbers should be generated separately and ensure they are independent of each other.

Step 2

Step2: Figure out the SIMULATION PROCESS (sequence of events matter)

2-1: general RN(RN1) for the #arrivals => determine #arrivals at each time slots (i.e. every 5 min)

2-2: for each arrival in (2-1) generate another RN (RN2) => determine whether it should be searched?

2-3: for each arrival that needs to be searched, generate the 3rd RN (RN3) to determine its search time

Implementation of Step 2

Time period	RN1 #ar-rivals	No.of Ar-rivals	RN2 Search?	Search? Yes or NO	RN3 Search Time	Search Time
12:00-12:05	61	2	15 94	Y N	42	4
12:05-12:10	04	0				
12:10-12:15	23	1	33	Y	02	2
12:15-12:20	59	2	46 19	N Y	47	4
12:20-12:25	85	3	48 52 96	N N N		
12:25-12:30	49	2	80 54	N N		

Implementation of Step 2 C'td

12:30-12:35	05	0				
12:35-12:40	17	1	23	Y	56	4
12:40-12:45	15	1	86	N		
12:45-12:50	69	2	69 44	N N		
12:50-12:55	72	2	11 32	Y Y	67 31	6 2
12:55-1:00	28	1	41	N		
TOTALS		17		6		22

Summary

What percentage of cars that arrive between 12:00 and 1:00 a.m are searched?

$6/17 = .3529$ or 35.29% according to the simulation.

What percentage of the time is the customs agent busy during the period from 12:00 and 1:00 a.m?

$22/60 = 36.67\%$

Critical Thinking:

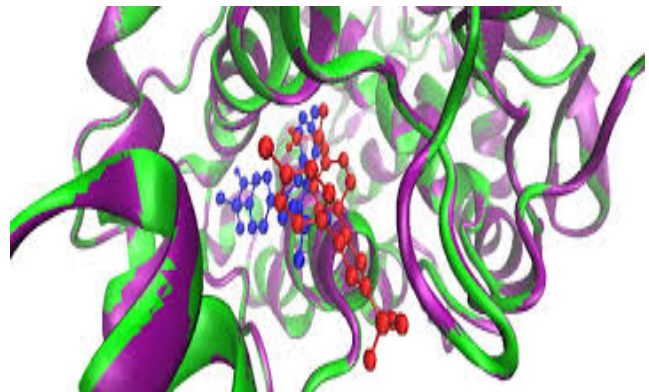
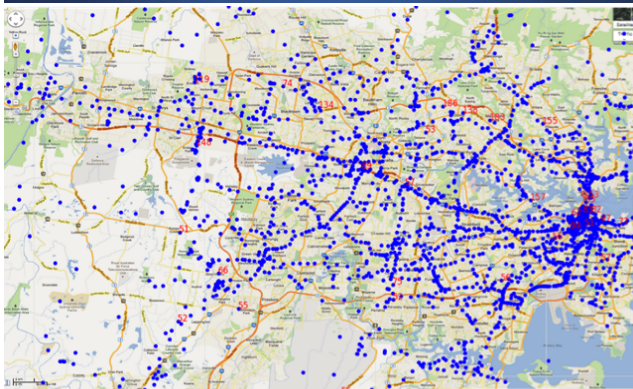
The expected number of arrivals from the probability distribution is:

$0(.10) + 1(.25) + 2(.45) + 3(.20) = 1.75$ cars/5-min

The simulation shows 17 arrivals in 60 minutes or $17/12 = 1.42$ cars/5-min.

Therefore the simulation underestimates the actual mean number of arrivals. We need to increase the scale of simulation later to ensure a more accurate result.

Applications of Simulation



Next Lecture

■ Multi-period Decision Analysis

